Exercise Sheet 16

Conventions are as in Lecture 16. For a Lie group G, $d_L x$ and $d_R x$ are the left and right Haar densities with the same value at e. The modular function is $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. We have

$$\int_G f(xg)\, d_L x = \Delta(g)^{-1} \int_G f(x)\, d_L x, \qquad d_R x = \Delta(x)^{-1}\, d_L x.$$

Some groups below are open subsets $U \subset \mathbb{R}^N$ with smooth group operations. For these, a density $m(x)\,dx$ with smooth $m > 0$ is left-invariant exactly when

$$m(gx)\left|\det J_g(x)\right| = m(x) \qquad (g, x \in U).$$

Here $J_g(x)$ is the Jacobian matrix of $x \mapsto gx$. Right invariance uses $x \mapsto xg$ instead. Exercise 16.6 is an optional extension.

Exercise 16.1. Let G be the $ax + b$ group of matrices

$$g(a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}, \qquad a > 0,\ b \in \mathbb{R}.$$

Use global coordinates (a, b). The Haar densities and the modular function are

$$d_L x = a^{-2}\, da\, db, \qquad d_R x = a^{-1}\, da\, db, \qquad \Delta(g(a,b)) = a^{-1}.$$

For a region E put

$$\mu_L(E) = \iint_E a^{-2}\, da\, db, \qquad \mu_R(E) = \iint_E a^{-1}\, da\, db.$$

Let $E = \{1 \le a \le 2,\ 0 \le b \le 1\}$, $g = g(2,0)$ and $h = g(1,3)$.

(a) Describe gE, Eg, hE and Eh. Compute their left measures and compare them with $\mu_L(E)$ and Δ.
(b) Compute $\mu_R(E)$, $\mu_R(gE)$ and $\mu_R(Eg)$.
(c) Describe $E^{-1} = \{x^{-1} : x \in E\}$ and show that $\mu_L(E^{-1}) = \mu_R(E)$.

Exercise 16.2. Let B be the group of matrices

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \qquad a, d > 0,\ b \in \mathbb{R}.$$

Use coordinates (a, b, d) on $(0,\infty)\times\mathbb{R}\times(0,\infty)$. Use the basis E_{11}, E_{12}, E_{22} of the Lie algebra.

(a) Compute the products gx' and $x'g$ for $g = (a,b,d)$ and $x' = (a',b',d')$. Find the Jacobian determinants of $x' \mapsto gx'$ and $x' \mapsto x'g$.
(b) Find the left and right Haar densities whose coefficient at the identity $(1,0,1)$ is 1.
(c) Compute Δ from the ratio of the densities. Check the result with $\det \mathrm{Ad}(g)$.
(d) For which g is $\Delta(g) = 1$? Is B unimodular?

Exercise 16.3. The group $\mathrm{GL}(n,\mathbb{R})$ is an open subset of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$, with coordinates x_{ij} and Lebesgue measure $dx = \prod dx_{ij}$. For $g \in \mathrm{GL}(n,\mathbb{R})$ let $\ell_g(X) = gX$ and $r_g(X) = Xg$ on $M_n(\mathbb{R})$.

(a) Show that $\det \ell_g = \det r_g = (\det g)^n$.

(b) Show that $|\det x|^{-n}\, dx$ is both left- and right-invariant.

(c) Compute $\det \mathrm{Ad}(g)$ directly and conclude that $\Delta \equiv 1$.

(d) What is the Haar density for $n = 1$? What is it on the subgroup $(0,\infty)$?

Exercise 16.4. For a connected Lie group, Lecture 16 gives $\Delta(\exp X) = e^{-\operatorname{tr}\operatorname{ad} X}$.

(a) Let N be the Heisenberg group of matrices

$$\begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}, \qquad x, y, z \in \mathbb{R},$$

with coordinates (x, y, z). Use the basis $X = E_{12}$, $Y = E_{23}$, $Z = E_{13}$. Compute the matrices of $\operatorname{ad} X$, $\operatorname{ad} Y$ and $\operatorname{ad} Z$. Check by Jacobians that $dx\,dy\,dz$ is left- and right-invariant.

(b) Use the basis H, E, F of $\mathfrak{sl}(2,\mathbb{R})$ with $[H,E] = 2E$, $[H,F] = -2F$ and $[E,F] = H$. Show that $\operatorname{tr}\operatorname{ad} X = 0$ for every X in the Lie algebra. You may use that $\mathrm{SL}(2,\mathbb{R})$ is connected. Conclude that $\mathrm{SL}(2,\mathbb{R})$ is unimodular.

(c) Let S be the group of matrices

$$\begin{pmatrix} e^{t} & 0 & u \\ 0 & e^{-t} & v \\ 0 & 0 & 1 \end{pmatrix}, \qquad t, u, v \in \mathbb{R},$$

with coordinates (t, u, v). Use the basis $T = E_{11} - E_{22}$, $U = E_{13}$, $V = E_{23}$. Show that S is unimodular. Confirm this by the Jacobians of left and right translation.

(d) For the $ax + b$ group, check that the formula gives $\Delta(g(e^t, 0)) = e^{-t}$.

Exercise 16.5. Let G be a Lie group. Let $L^2(G)$ be the completion of $C_c(G)$ for $\langle f, h\rangle = \int_G \overline{f}\, h\, d_L x$. Let

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = \Delta(g)^{1/2} f(xg).$$

You may use the following identity for $F \in C_c(G)$:

$$\int_G F(x^{-1})\, \Delta(x^{-1})\, d_L x = \int_G F(x)\, d_L x.$$

For $f \in C_c(G)$ define

$$(Jf)(x) = \Delta(x)^{-1/2} f(x^{-1}).$$

(a) Show that $\|Jf\| = \|f\|$.

(b) Show that $J^2 f = f$.

(c) Show that $J\lambda(g)f = \rho(g)Jf$.

(d) Conclude that λ and ρ are unitarily equivalent.

(e) Write J explicitly for the $ax + b$ group. Check the isometry in (a) there by substitution, using $d_L x = a^{-2}\, da\, db$ and $\Delta(g(a,b)) = a^{-1}$.

Exercise 16.6 (Optional extension). Let $\mathrm{Aff}(n)$ be the group of matrices

$$\begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}, \qquad A \in \mathrm{GL}(n,\mathbb{R}),\ v \in \mathbb{R}^n.$$

Write the elements as (A, v). The group is an open subset of $\mathbb{R}^{n^2+n}$.

(a) Compute $(A, v)(A', v')$ and the Jacobian determinants of left and right translation in (A', v').

(b) Find left and right Haar densities and the modular function.

(c) Compute $\det \mathrm{Ad}(A, v)$ on the Lie algebra pairs (X, w) with $X \in \mathfrak{gl}(n,\mathbb{R})$ and $w \in \mathbb{R}^n$. Compare with (b).

(d) Specialise to $n = 1$ and compare with the $ax + b$ group.